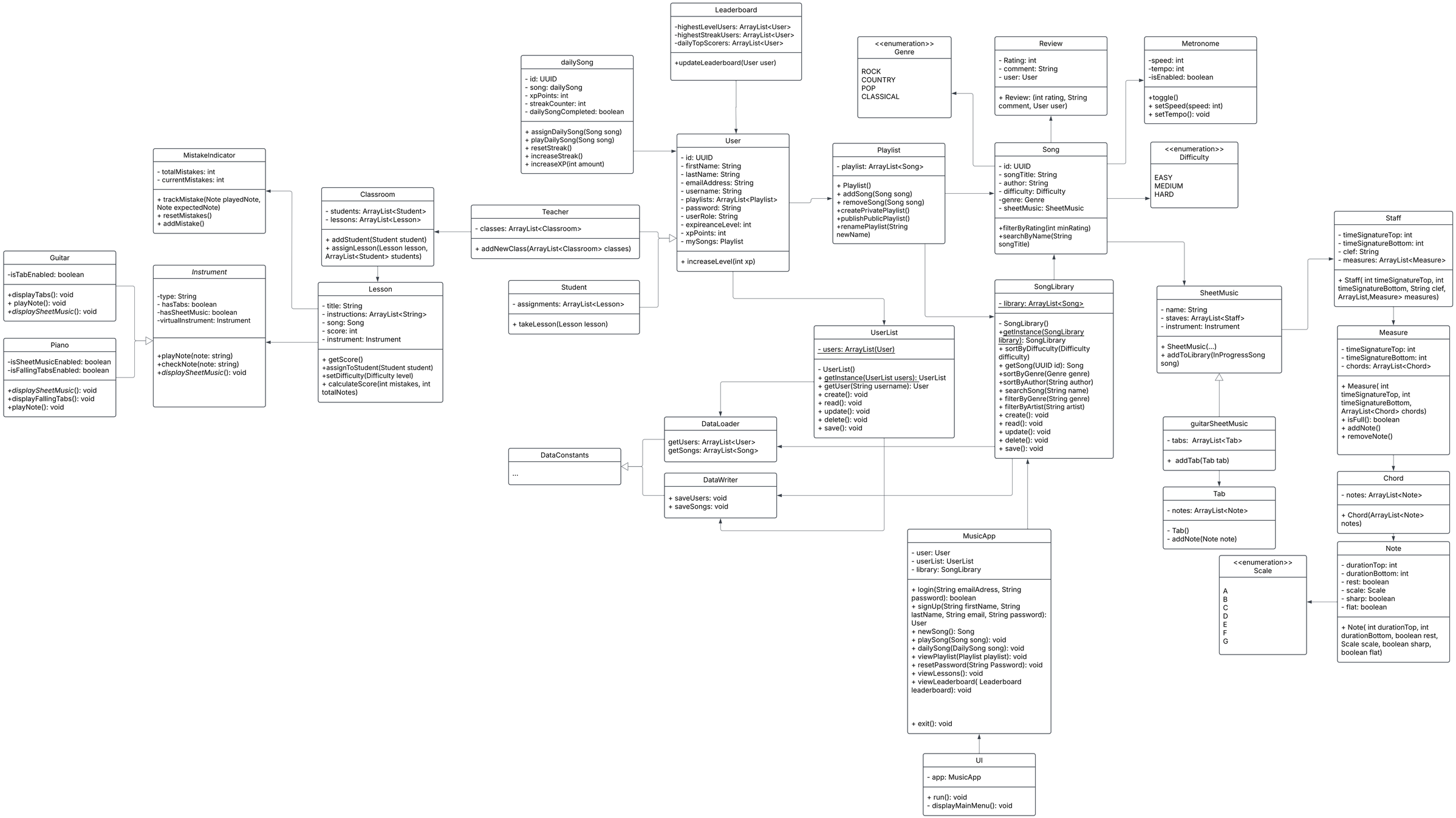

Leaderboard
-highestLevelUsers: ArrayList<User>
-highestStreakUsers: ArrayList<User>
-dailyTopScorers: ArrayList<User>
+updateLeaderboard(User user)
dailySong
- id: UUID
- song: dailySong
- xpPoints: int
- streakCounter: int
- dailySongCompleted: boolean
+ assignDailySong(Song song)
+ playDailySong(Song song)
+ resetStreak()
+ increaseStreak()
+ increaseXP(int amount)
<<enumeration>>
Genre
ROCK
COUNTRY
POP
CLASSICAL
Review
- Rating: int
- comment: String
- user: User
+ Review: (int rating, String comment, User user)
Metronome
-speed: int
-tempo: int
-isEnabled: boolean
+toggle()
+ setSpeed(speed: int)
+ setTempo(): void
MistakeIndicator
- totalMistakes: int
- currentMistakes: int
+ trackMistake(Note playedNote, Note expectedNote)
+ resetMistakes()
+ addMistake()
Classroom
- students: ArrayList<Student>
- lessons: ArrayList<Lesson>
+ addStudent(Student student)
+ assignLesson(Lesson lesson, ArrayList<Student> students)
Teacher
- classes: ArrayList<Classroom>
+ addNewClass(ArrayList<Classroom> classes)
User
- id: UUID
- firstName: String
- lastName: String
- emailAddress: String
- username: String
- playlists: ArrayList<Playlist>
- password: String
- userRole: String
- expireanceLevel: int
- xpPoints: int
- mySongs: Playlist
+ increaseLevel(int xp)
Playlist
- playlist: ArrayList<Song>
+ Playlist()
+ addSong(Song song)
+ removeSong(Song song)
+createPrivatePlaylist()
+publishPublicPlaylist()
+renamePlaylist(String newName)
Song
- id: UUID
- songTitle: String
- author: String
- difficulty: Difficulty
-genre: Genre
- sheetMusic: SheetMusic
+filterByRating(int minRating)
+searchByName(String songTitle)
<<enumeration>>
Difficulty
EASY
MEDIUM
HARD
Guitar
-isTabEnabled: boolean
+displayTabs(): void
+ playNote(): void
+displaySheetMusic(): void
Instrument
-type: String
- hasTabs: boolean
-hasSheetMusic: boolean
-virtualInstrument: Instrument
+playNote(note: string)
+checkNote(note: string)
+displaySheetMusic(): void
Piano
-isSheetMusicEnabled: boolean
-isFallingTabsEnabled: boolean
+displaySheetMusic(): void
+displayFallingTabs(): void
+playNote(): void
Lesson
- title: String
- instructions: ArrayList<String>
- song: Song
- score: int
- instrument: Instrument
+ getScore()
+assignToStudent(Student student)
+setDifficulty(Difficulty level)
+ calculateScore(int mistakes, int totalNotes)
Student
- assignments: ArrayList<Lesson>
+ takeLesson(Lesson lesson)
SongLibrary
- library: ArrayList<Song>
- SongLibrary()
+getInstance(SongLibrary library): SongLibrary
+ sortByDiffuculty(Difficulty difficulty)
+ getSong(UUID id): Song
+sortByGenre(Genre genre)
+sortByAuthor(String author)
+ searchSong(String name)
+ filterByGenre(String genre)
+ filterByArtist(String artist)
+ create(): void
+ read(): void
+ update(): void
+ delete(): void
+ save(): void
SheetMusic
- name: String
- staves: ArrayList<Staff>
- instrument: Instrument
+ SheetMusic(...)
+ addToLibrary(InProgressSong song)
Staff
- timeSignatureTop: int
- timeSignatureBottom: int
- clef: String
- measures: ArrayList<Measure>
+ Staff(int timeSignatureTop, int timeSignatureBottom, String clef, ArrayList,Measure> measures)
UserList
- users: ArrayList(User)
- UserList()
+ getInstance(UserList users): UserList
+ getUser(String username): User
+ create(): void
+ read(): void
+ update(): void
+ delete(): void
+ save(): void
DataLoader
getUsers: ArrayList<User>
getSongs: ArrayList<Song>
DataConstants
...
DataWriter
+ saveUsers: void
+ saveSongs: void
guitarSheetMusic
- tabs: ArrayList<Tab>
+ addTab(Tab tab)
Measure
- timeSignatureTop: int
- timeSignatureBottom: int
- chords: ArrayList<Chord>
+ Measure(int timeSignatureTop, int timeSignatureBottom, ArrayList<Chord> chords)
+ isFull(): boolean
+ addNote()
+ removeNote()
Tab
- notes: ArrayList<Note>
- Tab()
- addNote(Note note)
Chord
- notes: ArrayList<Note>
+ Chord(ArrayList<Note> notes)
<<enumeration>>
Scale
A
B
C
D
E
F
G
Note
- durationTop: int
- durationBottom: int
- rest: boolean
- scale: Scale
- sharp: boolean
- flat: boolean
+ Note(int durationTop, int durationBottom, boolean rest, Scale scale, boolean sharp, boolean flat)
MusicApp
- user: User
- userList: UserList
- library: SongLibrary
+ login(String emailAdress, String password): boolean
+ signUp(String firstName, String lastName, String email, String password): User
+ newSong(): Song
+ playSong(Song song): void
+ dailySong(DailySong song): void
+ viewPlaylist(Playlist playlist): void
+ resetPassword(String Password): void
+ viewLessons(): void
+ viewLeaderboard(Leaderboard leaderboard): void
+ exit(): void
UI
- app: MusicApp
+ run(): void
- displayMainMenu(): void